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\\USER\Investigators\Brady\TMSstudy_Feb2024\localizer_32ch_uncombined
TA: 14 sec Coil Selection: Manual Voxel Size: 0.5×0.5×7.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	250 mm
FoV Phase	100.0 %
Slice Thickness	7.0 mm
TR	8.6 ms
TE	4.00 ms
Averages	2
Concatenations	3
AutoAlign	---
Coil Elements	HEA;HEP

Contrast - Common

TR	8.6 ms
TE	4.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Each Measurement

Resolution - Common

FoV Read	250 mm
FoV Phase	100.0 %
Slice Thickness	7.0 mm
Base Resolution	256
Phase Resolution	91 %
Interpolation	On

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	250 mm
FoV Phase	100.0 %
Slice Thickness	7.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

Slice Group	3
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None

Physio - Signal

TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FoV Read	250 mm
FoV Phase	100.0 %
Phase Resolution	91 %
Dynamic Mode	Standard

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.00 ms
TR	8.6 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None

Sequence - Part 1

Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\Investigators\Brady\TMSstudy_Feb2024\AAHead_Scout_32ch
TA: 14 sec Coil Selection: Manual Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.2 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head
Coil Elements	HEA;HEP

Contrast - Common

TR	3.2 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.2 s

Resolution - Common

FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3

Resolution - Acceleration

Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Noise Masking	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.2 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Dynamic

Dynamic Mode	Standard
Flip Angle	8 deg
Measurements	1
Time to Center	6.2 s

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Save Original Images	On
Contrasts	1
TE	1.37 ms
TR	3.2 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\T1w_setter
TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	1st Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	8.0 mm
TR	11.0 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	BC

Contrast - Common

TR	11.0 ms
TE	4.80 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	8.0 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	8.0 mm
TR	11.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H40.0
L	0.0 mm
P	0.0 mm
H	40.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	50.000

Physio - Signal

1st Signal/Mode	None
TR	11.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	MPRAGE

\\USER\Investigators\Brady\TMSstudy_Feb2024\T1_MEMPRAGE_1.0mm_p4_vNav

TA: 4:09 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	2nd Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	1.0 mm
TR	2200.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	2200.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Magn. Preparation	Non-sel. IR
TI	1100 ms
Flip Angle	7 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	4
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FoV Read	256 mm
FoV Phase	100.0 %

Resolution - Common

Slice Thickness	1.0 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	4
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	1.0 mm
TR	2200.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	16.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	16.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	1100 ms
Dark Blood	Off
FoV Read	256 mm
FoV Phase	100.0 %
Phase Resolution	100 %
Dynamic Mode	Standard

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Magn. Preparation	Non-sel. IR
Save Original Images	On
Contrasts	4
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
TR	2200.0 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tfl_me
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Reordering	Linear
Bandwidth 1	650 Hz/Px
Bandwidth 2	650 Hz/Px
Bandwidth 3	650 Hz/Px
Bandwidth 4	650 Hz/Px
Echo Spacing	9.84 ms
Asymmetric Echo	Off
Turbo Factor	154

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav
Remeasure	20 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	RMS Only

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\Minn_CMRR_2.3mm_S8_rest_6min

TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0
Min. prep scans SB	0
Multi-band PE shift	0 1/FoV
MB kernel size	1
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\Minn_CMRR_2.3mm_S8_rest_6min

TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0

Sequence - Special

Min. prep scans SB	0
Multi-band PE shift	0 1/FoV
MB kernel size	1
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\Minn_CMRR_2.3mm_S8_rest_6min

TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0
Min. prep scans SB	0
Multi-band PE shift	0 1/FoV
MB kernel size	1
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\Minn_CMRR_2.3mm_S8_rest_6min

TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0

Sequence - Special

Min. prep scans SB	0
Multi-band PE shift	0 1/FoV
MB kernel size	1
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\CMRR_SEFieldMap_2.3mm_64sl_AP

TA: 8 sec Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	8000.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off

Resolution - Acceleration

Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard

System - Adjustments

B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8000.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	On
EPI Factor	90

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Fake MB factor for SB	1
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\CMRR_SEFieldMap_2.3mm_64sl_PA
TA: 8 sec Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	8000.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off

Resolution - Acceleration

Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard

System - Adjustments

B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8000.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	On
EPI Factor	90

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Fake MB factor for SB	1
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	On
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\localizer_32ch_uncombined
TA: 14 sec Coil Selection: Manual Voxel Size: 0.5×0.5×7.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	All Segments
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	250 mm
FoV Phase	100.0 %
Slice Thickness	7.0 mm
TR	8.6 ms
TE	4.00 ms
Averages	2
Concatenations	3
AutoAlign	---
Coil Elements	HEA;HEP

Contrast - Common

TR	8.6 ms
TE	4.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FoV Read	250 mm
FoV Phase	100.0 %
Slice Thickness	7.0 mm
Base Resolution	256
Phase Resolution	91 %
Interpolation	On

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	250 mm
FoV Phase	100.0 %
Slice Thickness	7.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P

Geometry - AutoAlign

Slice Group	3
Position	L0.0 A20.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L0.0 A20.0 H0.0
L	0.0 mm
A	20.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None

Physio - Signal

TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FoV Read	250 mm
FoV Phase	100.0 %
Phase Resolution	91 %
Dynamic Mode	Standard

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.00 ms
TR	8.6 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None

Sequence - Part 1

Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\Investigators\Brady\TMSstudy_Feb2024\AAHead_Scout_32ch

TA: 14 sec Coil Selection: Manual Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.2 ms
TE	1.37 ms
Averages	1
Concatenations	1
AutoAlign	Head
Coil Elements	HEA;HEP

Contrast - Common

TR	3.2 ms
TE	1.37 ms
Flip Angle	8 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.2 s

Resolution - Common

FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3

Resolution - Acceleration

Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Noise Masking	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	260 mm
FoV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.2 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A10.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Dynamic

Dynamic Mode	Standard
Flip Angle	8 deg
Measurements	1
Time to Center	6.2 s

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Save Original Images	On
Contrasts	1
TE	1.37 ms
TR	3.2 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\Minn_CMRR_2.3mm_S8_rest_6min

TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0

Sequence - Special

Min. prep scans SB	0
Multi-band PE shift	0 1/FoV
MB kernel size	1
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\Minn_CMRR_2.3mm_S8_rest_6min
TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0
Min. prep scans SB	0
Multi-band PE shift	0 1/FoV
MB kernel size	1
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\Minn_CMRR_2.3mm_S8_rest_6min
TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0
Min. prep scans SB	0
Multi-band PE shift	0 1/FoV
MB kernel size	1
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\Minn_CMRR_2.3mm_S8_rest_6min

TA: 6:09 min Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	Off
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
TE	34.80 ms
Averages	1
Multi-band accel. factor	8
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	650.0 ms
TE	34.80 ms
MTC	Off
Magn. Preparation	None
Flip Angle	50 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	554
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	650.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	8

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	650.0 ms
Multi-band accel. factor	8

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	554
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	Off
EPI Factor	90

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	5960 us
Min. prep scans	0

Sequence - Special

Min. prep scans SB	0
Multi-band PE shift	0 1/FoV
MB kernel size	1
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Sinc exc. pulse BWTP	5.20
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\CMRR_SEFieldMap_2.3mm_64sl_AP

TA: 8 sec Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	8000.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off

Resolution - Acceleration

Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard

System - Adjustments

B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8000.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	On
EPI Factor	90

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Fake MB factor for SB	1
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\CMRR_SEFieldMap_2.3mm_64sl_PA

TA: 8 sec Coil Selection: Manual Voxel Size: 2.3×2.3×2.3 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
TE	66.00 ms
Averages	1
Multi-band accel. factor	1
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	8000.0 ms
TE	66.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Disabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
Base Resolution	90
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	None
Advanced Reconstruction	Off

Resolution - Acceleration

Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	64
Distance Factor	0 %
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FoV Read	207 mm
FoV Phase	100.0 %
Slice Thickness	2.3 mm
TR	8000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	1

Geometry - AutoAlign

Slice Group	1
Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P8.0 H6.0
L	0.0 mm
P	8.0 mm
H	6.0 mm
Initial Orientation	T > C
T > C	-25.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard

System - Adjustments

B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.0 P8.0 H6.0 mm
Orientation	T > C-25.0
Rotation	0.00 deg
A >> P	207 mm
R >> L	207 mm
F >> H	148 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8000.0 ms
Multi-band accel. factor	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	Off
Temp. Highpass Filter	Off
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	2222 Hz/Px
Echo Spacing	0.62 ms
Free Echo Spacing	On
EPI Factor	90

Sequence - Part 2

Introduction	Off

Sequence - Special

Min. prep scans	0
Fake MB factor for SB	1
Delay before PC scans	0 us
SENSE1 coil combine	Off
Invert RO/PE polarity	On
Disable freq. update	Off
Suppress 16-bit DICOM	Off
Force equal slice timing	Off
FFT scale factor	1.00
Fat saturation FA	110.00 deg
Fat sat. offset	0.00 Hz
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\USER\Investigators\Brady\TMSstudy_Feb2024\T1w_setter

TA: 1 sec Coil Selection: Manual Voxel Size: 8.0×8.0×8.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	1st Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	8.0 mm
TR	11.0 ms
TE	4.80 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	BC

Contrast - Common

TR	11.0 ms
TE	4.80 ms
MTC	Off
Flip Angle	2 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	8.0 mm
Base Resolution	32
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Slice Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	32
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	8.0 mm
TR	11.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 P0.0 H40.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H40.0
L	0.0 mm
P	0.0 mm
H	40.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto

System - Adjustments

Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	50.000

Physio - Signal

1st Signal/Mode	None
TR	11.0 ms
Concatenations	1

Sequence - Part 1

Sequence Name	ABCD
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	4882 Hz/Px
Echo Spacing	0.27 ms
Free Echo Spacing	Off
EPI Factor	32

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

Protocol filename	MPRAGE

\\USER\Investigators\Brady\TMSstudy_Feb2024\T1_MEMPRAGE_1.0mm_p4_vNav

TA: 4:09 min Coil Selection: Manual Voxel Size: 1.0×1.0×1.0 mm³ Acc:: 4 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	2nd Segment
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	1.0 mm
TR	2200.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain
Coil Elements	HEA;HEP

Contrast - Common

TR	2200.0 ms
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
Magn. Preparation	Non-sel. IR
TI	1100 ms
Flip Angle	7 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	4
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FoV Read	256 mm
FoV Phase	100.0 %

Resolution - Common

Slice Thickness	1.0 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	4
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FoV Read	256 mm
FoV Phase	100.0 %
Slice Thickness	1.0 mm
TR	2200.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	16.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Manual
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	16.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.235492 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2200.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	Non-sel. IR
TI	1100 ms
Dark Blood	Off
FoV Read	256 mm
FoV Phase	100.0 %
Phase Resolution	100 %
Dynamic Mode	Standard

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Magn. Preparation	Non-sel. IR
Save Original Images	On
Contrasts	4
TE 1	1.69 ms
TE 2	3.55 ms
TE 3	5.41 ms
TE 4	7.27 ms
TR	2200.0 ms

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off

Sequence - Part 1

Sequence Name	tfl_me
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Reordering	Linear
Bandwidth 1	650 Hz/Px
Bandwidth 2	650 Hz/Px
Bandwidth 3	650 Hz/Px
Bandwidth 4	650 Hz/Px
Echo Spacing	9.84 ms
Asymmetric Echo	Off
Turbo Factor	154

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off

Sequence - Special

Readout polarity	Positive
Nav. location	Before
Apply moco to	parent and nav
Remeasure	20 TRs
Reacq. threshold	0.50
Feedback Delay	80 ms
Moco ref. image	Use Temp Ref
K-space streaming	None
ABCD navigator	On
Add. grad time	0.00 ms
Apply freq to	parent and nav
Averaging	RMS Only

Sequence - Assistant

SAR Assistant	Off